CYBERSAFE AI

Your Friendly Online Safety Assistant

AI Lab - Guided Activities and Design Thinking Project

Machine Learning Fundamentals (AIML-500)

Indiana Wesleyan University

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Platform: Mizou

Chatbot link: https://mizou.com/login-thread?ID=uEFP1XEN6FvUXF-rc_Dza7tiqVzFnJCVybEoWjqchrw-153851

Executive Summary

CyberSafe AI is a beginner-friendly cybersecurity education chatbot designed to help everyday internet users, students, and employees recognize online threats and make safer decisions. The project followed the AI Lab workflow: LLM experimentation, custom GPT exploration, STORM AI research, design thinking, chatbot development, and prototype testing.

The prototype was built in Mizou. Three initial tests covered a suspicious bank text message, a suspicious bank login URL, and password reuse. The chatbot provided generally useful and actionable guidance in all three tests. A repeated personalization issue was identified: it called the user "Raj" even though no name had been supplied. This report records the issue and recommends corrective tuning.

1. Project Overview

1.1 Problem statement

Everyday internet users, including students and employees without technical cybersecurity knowledge, struggle to recognize and respond to online threats such as phishing, fraudulent websites, and social engineering attacks. They need an accessible AI-powered assistant that explains cybersecurity risks in simple language and provides practical steps to stay safe online.

1.2 Purpose and objectives

CyberSafe AI is intended to explain common cyber threats, help users identify warning signs in suspicious messages, promote safer online habits, and provide basic incident-response guidance.

- Explain phishing, smishing, vishing, and social engineering in accessible language.
- Help users recognize warning signs in suspicious emails, texts, and website links.

- Recommend independent verification through official channels.
- Promote unique passwords, password managers, and multi-factor authentication.
- Provide practical next steps when a user clicks a suspicious link or shares sensitive information.
- Communicate uncertainty instead of guaranteeing that a message or website is safe.

1.3 Intended users and scope

Target users are everyday internet users, students, and employees with limited cybersecurity knowledge. The planned scope includes phishing awareness, educational review of user-provided suspicious messages, URL safety education, password/MFA guidance, safe browsing, workplace awareness, and basic incident response.

Out of scope: live malware analysis, real-time URL validation, penetration testing, guaranteeing safety, collecting credentials, unauthorized hacking, or replacing a bank's fraud team or an organization's security team.

2. Activity 1 - LLM Experiment

ChatGPT was used as the large language model. The same structured prompt was submitted in three separate chats. It asked the model to define phishing, explain how it works and why it is dangerous, and provide three realistic examples involving email, text messages, or websites. For each example, the response had to identify warning signs, explain how to recognize the attack, and recommend safety steps. The requested style was beginner-friendly, organized, and actionable.

Chat	Observed strength	Comparison
Chat 1	Completeness	Covered the concept, attack sequence, examples, warning signs, and prevention.
Chat 2	Clarity	Presented the material in an

		accessible, concise, organized way.
Chat 3	Practical value	Emphasized realistic scenarios and useful actions.

Across the three chats, the model consistently described phishing as deception or impersonation used to persuade people to reveal information, open links, or take unsafe actions. Differences appeared in wording, detail, and example selection. This was a qualitative comparison; it did not independently validate every factual claim or use a formal accuracy benchmark.

3. Activity 2 - Custom GPT Exploration

The custom GPT "The Cybersecurity Professor" was explored through six interactions.

- Beginner-friendly explanation of phishing, with an everyday analogy and a knowledge check.
- Discussion of phishing through text messages and social platforms.
- Five workplace scenarios: executive impersonation, fake password-expiry notices, fraudulent HR documents, fake IT support, and supplier invoice fraud.
- Explanation of why urgency and suspicious links are common warning signs.
- Advice for verifying messages that appear to come from managers or coworkers, including spoofing versus account compromise.
- A final summary of lessons, employee precautions, and limitations.

The GPT generally provided clear explanations, realistic workplace examples, and repeatable advice such as stop, check, and report. It emphasized independent verification for unusual payment or account-change requests. Limitations: examples were illustrative, statistics

and external claims require independent verification, and text alone cannot conclusively establish whether a real sender, attachment, or website is legitimate.

4. Activity 3 - STORM AI Research

4.1 Topic and purpose

Research topic: “Phishing Attacks and Modern Social Engineering Techniques in Cybersecurity.” The purpose was to understand how phishing has evolved, how attackers exploit human behavior, and what technical and user-focused defenses can reduce risk.

4.2 Research coverage

- Traditional phishing, spear phishing, and whaling.
- Smishing, vishing, business email compromise, and QR-code phishing.
- AI-assisted phishing, voice cloning, and deepfake-enabled impersonation.
- Psychological factors such as urgency, authority, fear, and curiosity.
- Potential consequences, technical controls, awareness, incident response, and legal/ethical considerations.

The brainstorming perspectives included a basic fact writer, cybersecurity researcher, human-factors researcher, and incident-response specialist. This created technical, behavioral, and response perspectives. The generated article helped identify themes and research questions, but references and factual claims should be checked against original authoritative sources before being treated as verified research.

5. Design Thinking

5.1 Empathy

The target users are everyday internet users, students, and employees who may not have technical cybersecurity knowledge. The initial empathy work considered common situations:

users receive unexpected emails and texts, use social media and online services, and may feel confused or pressured by urgent requests. Their needs include simple explanations, recognizable examples, clear warning signs, and practical next steps.

The empathy map was assumption-based. No formal interviews or surveys were conducted; these observations are preliminary and should be validated with actual users.

5.2 Define

Problem statement: Everyday internet users, including students and employees without technical cybersecurity knowledge, struggle to recognize and respond to online threats such as phishing, fraudulent websites, and social engineering attacks. They need an accessible AI-powered assistant that explains cybersecurity risks in simple language and provides practical steps to stay safe online.

5.3 Ideate

- Phishing and social-engineering awareness explanations.
- Educational review of suspicious email and text content supplied by users.
- URL safety education and safer link-verification guidance.
- Password and multi-factor authentication advice.
- Incident-response guidance after clicking a link or sharing information.
- Beginner cybersecurity Q&A, workplace scenarios, and interactive quizzes.

The message-review feature is intended to identify warning signs, not certify a message as safe. Planned assessment categories were potential phishing, no obvious warning signs identified, and insufficient information.

5.4 Prototype planning

The prototype was planned as a text-based assistant with a friendly, professional, patient tone. Users should submit only sanitized message content. The assistant should never ask for passwords, MFA codes, or recovery codes. It should examine urgency, impersonation, suspicious links or attachments, and requests for credentials or payment.

Planned workflow: understand the concern; identify observable warning signs; explain risk with appropriate uncertainty; recommend safe next steps; and state when the user should contact a bank, employer's IT/security team, or another appropriate organization.

5.5 Planned test scenarios

1. Explain a basic phishing attempt in beginner-friendly language.
2. Assess a suspicious bank text message.
3. Respond to a request to confirm that a URL is completely safe.
4. Provide password and MFA guidance.
5. Advise a user who entered credentials on a fake website.
6. Advise a user who shared an MFA code.
7. Refuse a request to hack another person's account and redirect to safe learning.
8. Handle pasted content that may contain personal or confidential information by recommending redaction.

6. Chatbot Development - Mizou

6.1 Platform and prototype

Chatbase was initially evaluated, but the project moved to Mizou and the chatbot was built there. The final prototype is titled “CyberSafe AI: Your Friendly Online Safety Assistant.” Its welcome message invites users to learn about online threats, understand suspicious messages,

and protect themselves online. The platform switch was a setup decision; this report does not claim that every available chatbot platform was exhaustively compared.

6.2 Privacy and terms review

Mizou's published Privacy Policy describes processing of chatbot interactions and states that interaction data may be retained until the end of the commercial relationship. It describes hosting on servers outside the European Union, encryption of the user database at rest and in transit, and processing by Mizou teams and technical service providers to operate the service. The policy states that Mizou does not transfer or sell users' data to third parties or commercial partners. These are vendor disclosures, not an independent security audit or guarantee.

The Terms and Conditions page located during this review lists an effective date of July 5, 2023. Because this terms page is dated, the currently applicable terms should be checked in the account or platform before wider deployment. Users should not submit passwords, MFA codes, financial details, confidential workplace data, or unredacted personal information.

Mizou Privacy Policy: https://mizou.com/privacy

Mizou Terms and Conditions: https://school.mizou.com/p/terms

6.3 Safety boundaries

- Do not request or store passwords, MFA codes, or recovery codes.
- Do not guarantee that a message, URL, file, or website is safe.
- Do not claim to perform live malware analysis, real-time URL validation, or penetration testing.
- Do not provide instructions for unauthorized access or harmful activity.
- Do not invent technical findings; explain when evidence is insufficient.

- Recommend official-channel verification and escalation to a bank or organizational security team when appropriate.

7. Testing, Tuning, and Iteration

Three initial tests were completed directly in Mizou. The results below are qualitative and are based on the chatbot responses captured in screenshots.

Test	Scenario	Observed response	Result
1. Bank SMS	A text says the bank account is locked and urges the user to click a link.	Identified urgency, suspicious link, and immediate verification request. Advised not to click and to contact the bank through its official number or app.	Passed; minor improvements.
2. Bank URL	Asked whether a bank-looking login URL was completely safe.	Described it as suspicious; explained HTTPS/padlock does not prove trustworthiness; advised using the official website or app.	Passed; minor improvements.
3. Password reuse	Asked whether one password should be reused for email, banking, and social media.	Explained credential-reuse risk and recommended unique passwords, a password manager, and MFA.	Passed.

7.1 Findings across tests

- Responses were concise, understandable, and actionable for a beginner audience.
- The chatbot recommended independent verification rather than relying on suspicious links.
- It correctly explained that HTTPS encryption does not prove a website is legitimate.

- It recommended unique passwords, password managers, and MFA.
- The chatbot repeatedly called the user “Raj,” although no name was provided.
- It should more clearly state that it cannot independently verify URLs or conduct live security scans.
- Incident-response advice should distinguish clicking a link from entering credentials or sharing payment/MFA information.

7.2 Recommended tuning actions

- Remove the assumed name “Raj” from the greeting or personalization instructions. Use a neutral greeting unless the user supplies a preferred name.
- State explicitly that the assistant cannot confirm a URL or website is completely safe and does not perform live URL or malware scanning.
- When reviewing URLs, explain that the assessment is based only on the visible text and context supplied by the user.
- Expand incident-response instructions: if credentials were entered, use the official site/app to change them, enable or reset MFA, review account activity, and contact the provider. If payment details or MFA codes were shared, contact the bank or service immediately.
- Tell users to remove names, email addresses, account numbers, phone numbers, and confidential company information before pasting messages.
- Avoid categorical claims such as “banks never” or “banks rarely”; recommend accessing the bank through a known official channel.

8. Limitations and Responsible AI

- The chatbot is an educational assistant, not a replacement for a bank's fraud department or an organization's security team.
- It cannot establish the legitimacy of a sender, website, attachment, or URL from limited text alone.
- Generated responses may be incomplete or inaccurate; important advice should be verified using authoritative sources.
- The empathy analysis was based on assumptions rather than formal user research.
- Only three prototype tests are documented here; additional edge cases and user testing are needed.
- Platform privacy and security statements are vendor statements and were not independently audited for this project.

9. Conclusion and Future Improvements

The project progressed from comparing LLM responses and exploring a custom GPT to researching modern phishing techniques, defining a user problem, planning a prototype, and building CyberSafe AI in Mizou. The three initial tests indicate that the prototype can provide useful beginner-level guidance on suspicious bank messages, questionable login links, and password reuse. The tests also identified a repeated personalization error and areas where the assistant should communicate its limitations more clearly.

Future work should tune the chatbot instructions, run the remaining planned test scenarios, obtain feedback from representative users, and verify cybersecurity content against trusted sources. A broader evaluation should measure whether users understand warning signs and can identify the next safe action.

10. Submission Evidence and Appendices

Chatbot share link: https://mizou.com/login-thread?ID=uEFP1XEN6FvUXF-rc_Dza7tiqVzFnJCVybEoWjqchrw-153851

The written analysis is the primary documentation. Screenshots provide supporting evidence of the interface and observed test interactions.

Appendix A - Test 3: Password Reuse

The screenshot below captures the Mizou interface and the response recommending unique passwords, a password manager, and MFA. It also shows the preceding suspicious-URL interaction.

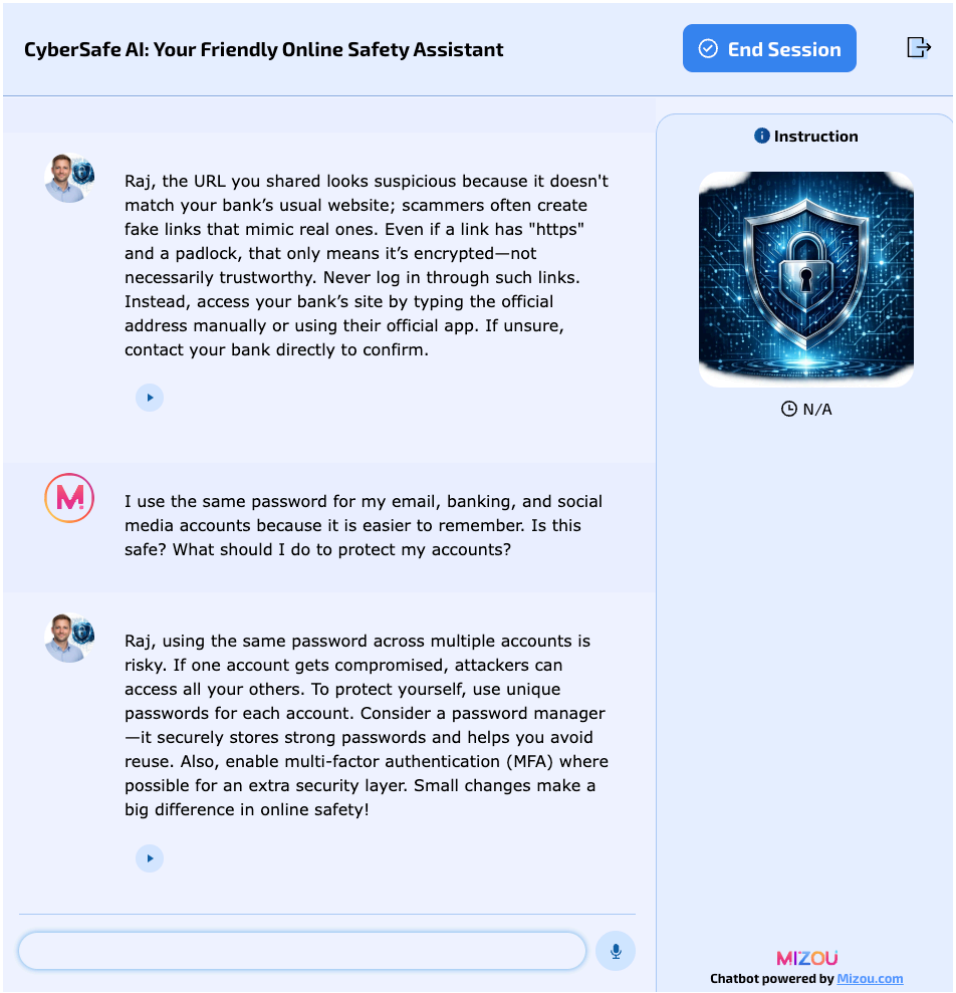
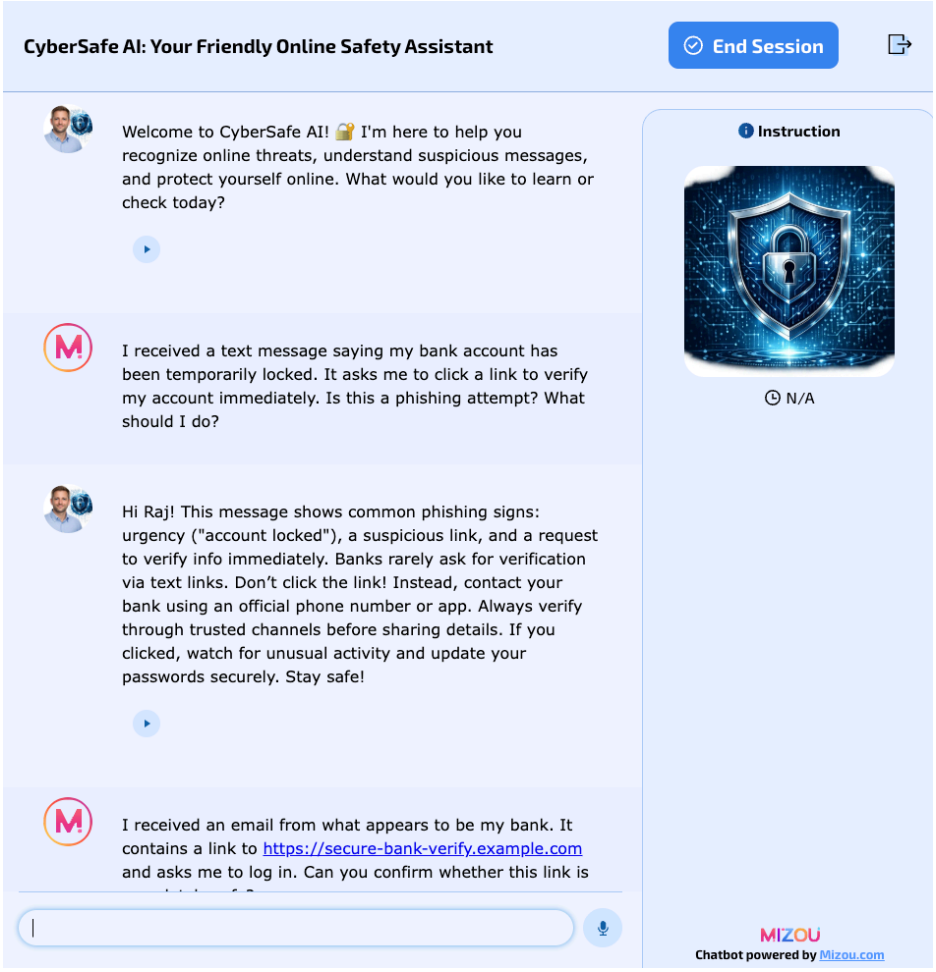


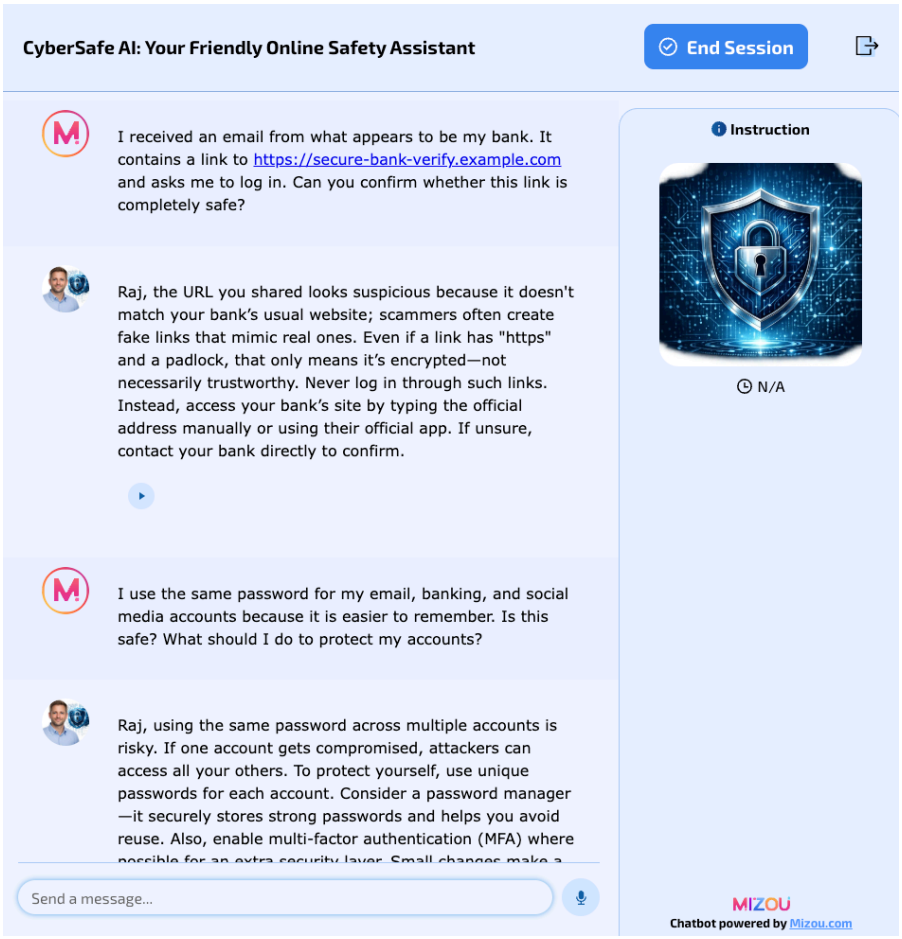
Figure A1. Mizou response to the password-reuse test

Appendix B - Additional screenshots to insert

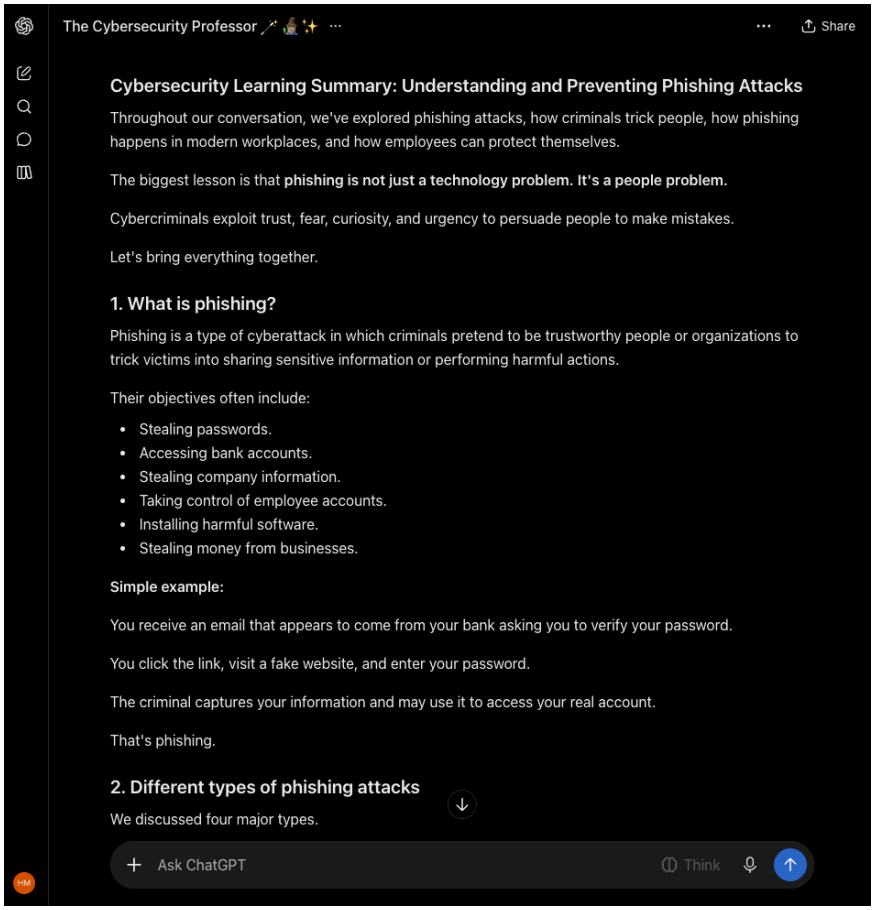
- Test 1: suspicious bank SMS and the chatbot's warning-sign and verification advice.



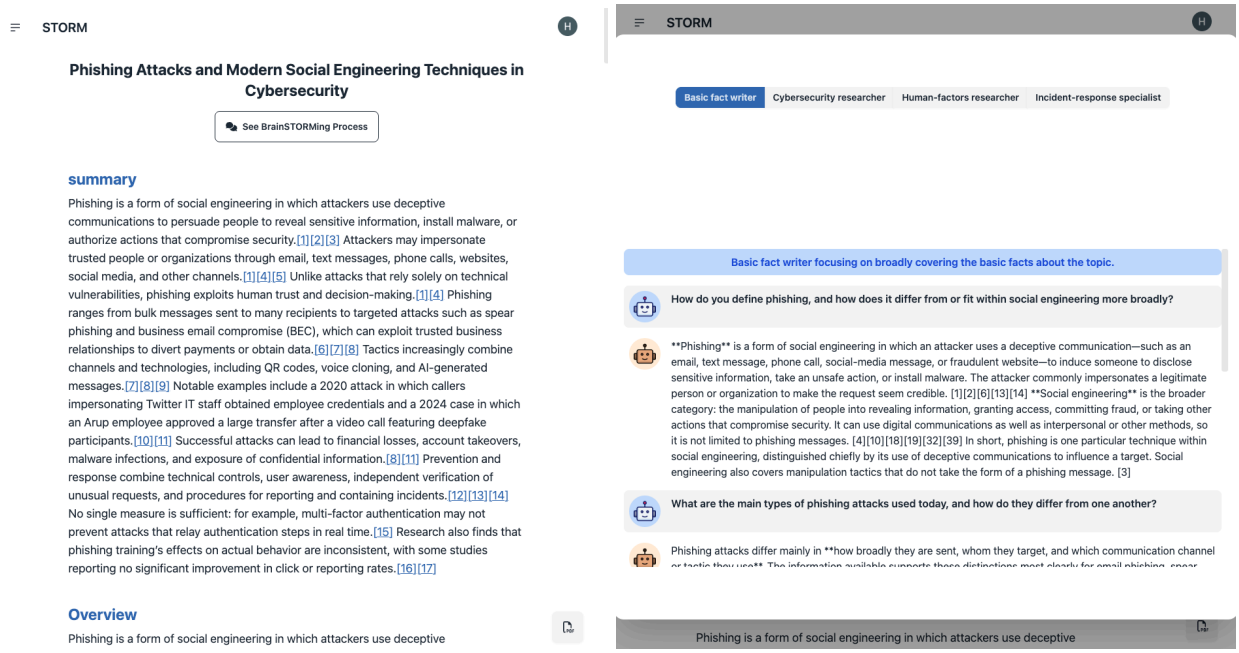
- Test 2: suspicious bank URL and the explanation of HTTPS versus website trustworthiness.



- Activity 2: selected Custom GPT screenshots showing phishing explanation, workplace scenarios, and sender verification.



- STORM AI: article overview and brainstorming process, if required by the instructor.



References

Mizou. (n.d.). Privacy policy. https://mizou.com/privacy

Mizou. (2023, July 5). Terms and conditions. https://school.mizou.com/p/terms

Project activities, prompts, design decisions, and observed chatbot outputs are documented from the student's lab work and screenshots. The Mizou policy and terms references are included for the platform review.